Prompting Substitutes for Scale on Reasoning, but Not on Knowledge

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DRAFT — GENERATED FROM VERIFIED RESULTS · SEEDS 42 & 43, N = 600 / CONDITION

Abstract. Practitioners choosing how to spend a limited budget face a recurring question: invest in a larger model, or in better prompting? We study this trade-off in a controlled setting, sweeping two instruction-tuned model families — Qwen2.5 (0.5B–7B) and Llama-3 (1B–8B), 4-bit quantization held constant — across four prompting strategies (zero-shot, few-shot, chain-of-thought, and structured JSON output) on three tasks spanning sentiment (SST-2), knowledge (MMLU), and multi-step arithmetic (GSM8K). We quantify prompting as a **parameter-equivalent exchange rate**: how much model scaling a strategy substitutes for. Contrary to the intuition that prompting helps least where models are weakest, we find the exchange rate is **largest on reasoning and smallest on knowledge**. On GSM8K, chain-of-thought (CoT) at a given size exceeds zero-shot accuracy at a much larger size — a Qwen-1.5B with CoT (55.7%) outperforms a Qwen-7B without it (14.3%), and a Llama-3B with CoT (74.7%) beats a Llama-8B without it (22.7%). On MMLU, no strategy improves on zero-shot by more than a few points; accuracy is governed almost entirely by parameter count. Both effects replicate across families. A third effect is model-specific: in Qwen, CoT *degrades* multiple-choice knowledge QA by 6–17pp (worsening with scale), whereas in Llama CoT is neutral-to-helpful — so we report it as family-dependent, not universal. Robustness checks (two random seeds, a continuous numeric-presence metric, and re-extraction from full outputs) confirm these effects are not scoring artifacts. All experiments run on a single 16GB laptop; code and prompts are released at github.com/ssamalsamir/prompting-vs-model-scaling.

1. Introduction

Deploying a language model means choosing both *which model* and *how to prompt it*, and these choices compete for the same budget: a larger model costs compute, memory, and latency, while prompt engineering is nearly free but bounded in effect. Prior work establishes the two levers separately — that few-shot ability emerges with scale (Brown et al., 2020), that chain-of-thought helps larger models reason (Wei et al., 2022b; Kojima et al., 2022), and that accuracy follows smooth scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) — but rarely quantifies how the two trade off against each other on a single, controlled model family.

We ask two questions. **RQ1 (exchange rate):** how much model scaling does each prompting strategy substitute for, and does this depend on task type? **RQ2 (CoT's cost):** does the strategy that helps reasoning carry over to other task types, or can it hurt?

Our central finding inverts a natural intuition. One might expect prompting to matter most for small, weak models and wash out as scale takes over. Instead, **prompting substitutes for scale on reasoning but not on knowledge**: a good prompt buys several model tiers on GSM8K yet almost nothing on MMLU, where facts must live in the parameters.

Contributions. (1) A controlled, statistically-tested measurement of the prompt–parameter exchange rate across two model families and three task types, with two random seeds (n = 600 per condition), Wilson confidence intervals, and Holm-adjusted McNemar tests. (2) Evidence that the exchange rate is taskdependent and prompt-specific — large for CoT on arithmetic, near-zero for any prompt on knowledge. (3) A demonstration that CoT can be double-edged — unlocking arithmetic while degrading multiple-choice knowledge QA — but that this degradation is *model-family-dependent*, verified against extraction and metric artifacts. (4) A reproducible single-laptop harness.

2. Related Work

Prompting and scale. In-context few-shot learning emerges with model size (Brown et al., 2020); chain-of-thought prompting improves reasoning primarily in larger models (Wei et al., 2022b; Kojima et al., 2022). The interaction between few-shot benefit and model size has been studied directly (San Joaquin & Haroen, 2022), and recent work shows small models struggle to exploit long reasoning chains (Li et al., 2025). None of these defines a quantitative prompt-as-parameters exchange rate on a controlled same-family ladder.

Scaling and emergence. Performance follows power-law scaling (Kaplan et al., 2020; Hoffmann et al., 2022). Some abilities appear abruptly with scale (Wei et al., 2022a), though this "emergence" may partly be an artifact of discontinuous metrics (Schaeffer et al., 2023) — a concern we address directly in §4.4.

Format and prompt sensitivity. Enforced output formats can degrade reasoning (Tam et al., 2024); meaning-preserving format changes swing accuracy substantially and the sensitivity persists across scale (Sclar et al., 2023); few-shot example ordering alone can dominate performance (Lu et al., 2022). GSM8K itself may reflect brittle pattern-matching (Mirzadeh et al., 2024). Our closest points of comparison are Tam et al. (2024), who study format restrictions but across a heterogeneous set of models rather than a controlled size ladder, and San Joaquin & Haroen (2022); neither frames results as an exchange rate nor contrasts knowledge against multi-step reasoning as we do.

3. Methods

Models. Two instruction-tuned families swept across parameter count with 4-bit quantization held constant, so size is the only within-family variable: Qwen2.5-Instruct at 0.5B / 1.5B / 3B / 7B and Llama-3 at 1B / 3B / 8B. We run locally via MLX on an Apple M4 (16GB), one model resident at a time.

Tasks. SST-2 (binary sentiment), MMLU (4-way multiple-choice knowledge, stratified across 57 subjects), GSM8K (grade-school multi-step arithmetic). n = 300 examples per condition. Both families were evaluated at two seeds (42 and 43); per-condition results are pooled over seeds (n = 600) in the confidence intervals reported in Appendix C.

Strategies. Zero-shot; few-shot (fixed hand-written exemplars); CoT (reasoning followed by a parsed final answer); structured (JSON-only output). Full templates in Appendix A.

Decoding and scoring. Greedy decoding (temperature 0), deterministic. Task-specific answer extraction: numeric match for GSM8K, exact label/letter match otherwise.

Statistics. Accuracy with 95% Wilson score intervals (≈ ±5–6pp at n = 300); paired McNemar tests (continuity-corrected, Holm-adjusted) for within-model strategy comparisons. We define the **exchange rate** operationally: prompting at size S "matches" scaling if the best-prompt accuracy at S meets or exceeds zero-shot accuracy at the next-larger size.

Robustness. To address the concern that exact-match exaggerates discontinuities (Schaeffer et al., 2023), for GSM8K we additionally record *numeric-present* (whether the correct value appears anywhere in the output) and the relative error of the committed answer; and we re-extract answers from full (untruncated) outputs (§4.4).

4. Results

4.1 Main results

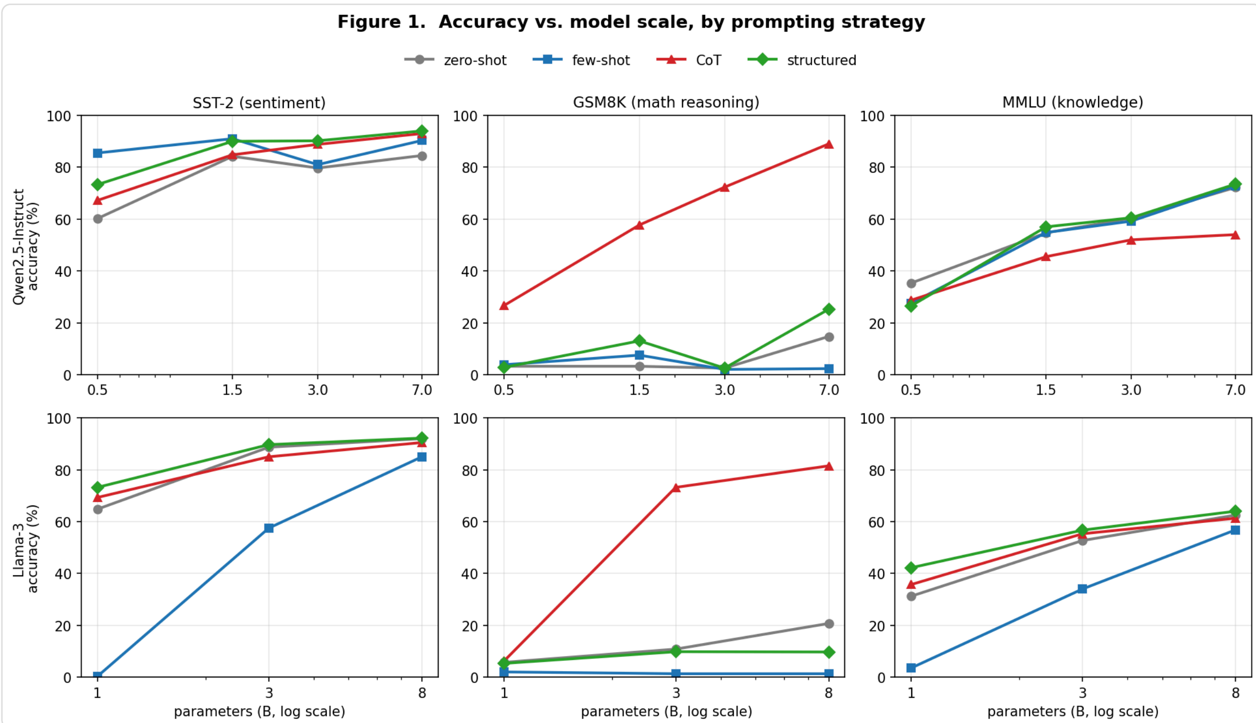

Figure 1. Accuracy vs. model scale (log axis) for each prompting strategy, by family (rows) and task (columns). On GSM8K, the CoT curve (red) separates sharply from all others; on MMLU the strategies bunch together and rise mainly with scale; on SST-2 the spread is moderate. Note Llama-3's erratic few-shot curve (blue).

Task	Model	zero-shot	few-shot	CoT	structured
SST-2	Qwen-0.5B	62.0	**85.3**	69.7	72.0
	Qwen-1.5B	86.7	**92.7**	84.3	91.0
	Qwen-3B	80.7	81.7	89.0	**90.7**
	Qwen-7B	87.3	91.0	94.7	**95.3**
MMLU	Qwen-0.5B	**32.7**	26.3	25.3	25.0
	Qwen-1.5B	57.3	55.3	47.3	**60.0**
	Qwen-3B	59.7	59.7	49.3	**60.0**
	Qwen-7B	72.3	**74.0**	55.3	73.7
GSM8K	Qwen-0.5B	4.3	4.0	**25.3**	2.3
	Qwen-1.5B	2.3	8.3	**55.7**	14.3
	Qwen-3B	2.3	1.7	**70.7**	2.7
	Qwen-7B	14.3	2.7	**88.0**	24.0

Table 1. Qwen2.5 accuracy (%). Green = best strategy for that row; red = CoT underperforms zero-shot.

Task	Model	zero-shot	few-shot	CoT	structured
SST-2	Llama-1B	67.3	0.7	71.0	**72.0**
	Llama-3B	89.3	60.0	86.0	**90.0**
	Llama-8B	**92.0**	85.0	90.3	91.7
MMLU	Llama-1B	31.3	4.0	37.3	**38.3**
	Llama-3B	52.3	35.0	56.3	**57.3**
	Llama-8B	60.7	55.0	58.7	**62.3**
GSM8K	Llama-1B	6.3	1.3	**7.3**	6.7
	Llama-3B	10.0	1.3	**74.7**	10.7
	Llama-8B	22.7	1.0	**79.0**	9.7

Table 2. Llama-3 accuracy (%). Note few-shot's collapse (red) and that CoT does *not* underperform zero-shot on MMLU here — the key cross-family divergence.

4.2 The exchange rate is large on reasoning, near-zero on knowledge (RQ1)

Figure 3. The prompt-parameter exchange rate on GSM8K: a small model with CoT beats a much larger one without it

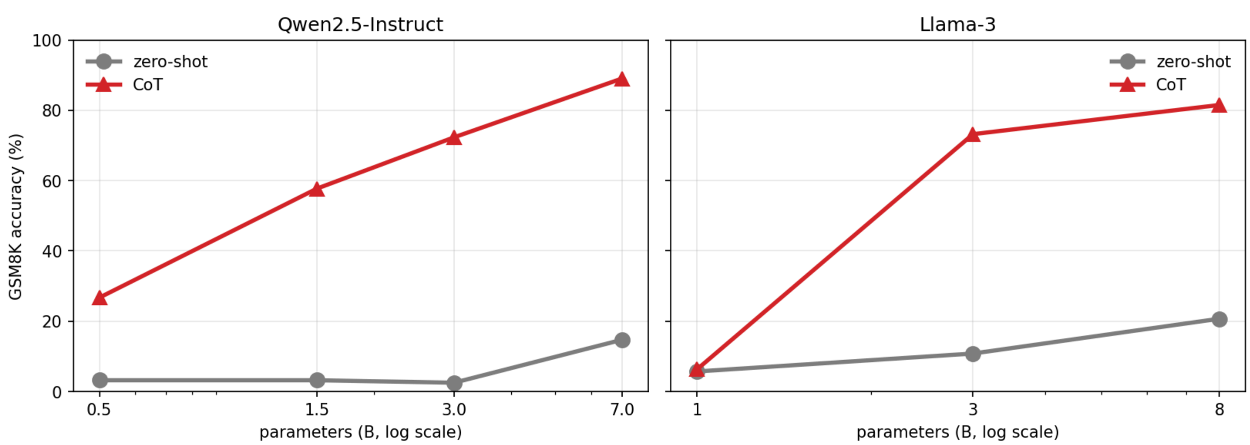

Figure 3. Zero-shot vs. CoT accuracy on GSM8K. In both families, CoT at a small size sits far above zero-shot at a much larger size — e.g. Qwen-1.5B + CoT (55.7%) exceeds Qwen-7B zero-shot (14.3%); Llama-3B + CoT (74.7%) exceeds Llama-8B zero-shot (22.7%).

GSM8K — prompting » scaling. CoT at every size exceeds zero-shot at the next size up. A 1.5B Qwen with CoT outperforms a 7B Qwen without it by 41pp — prompting worth roughly a 5× parameter increase — and the pattern repeats in Llama. Crucially this is CoT-specific: few-shot and structured remain near the floor, because answer-only or format-constrained prompts suppress the reasoning the task requires. (Llama-1B is an instructive floor: too weak to reason even with CoT.)

MMLU — scaling is irreplaceable. The best strategy improves on zero-shot by at most a few points (≤2.7pp in Qwen, ≤7pp in Llama, shrinking with scale). Accuracy rises with parameters regardless of prompt. Knowledge either resides in the weights or it does not.

SST-2 — prompting helps moderately, with best-strategy gains of 6–23pp, often equivalent to about one model tier.

4.3 CoT is double-edged — but the cost is model-specific (RQ2)

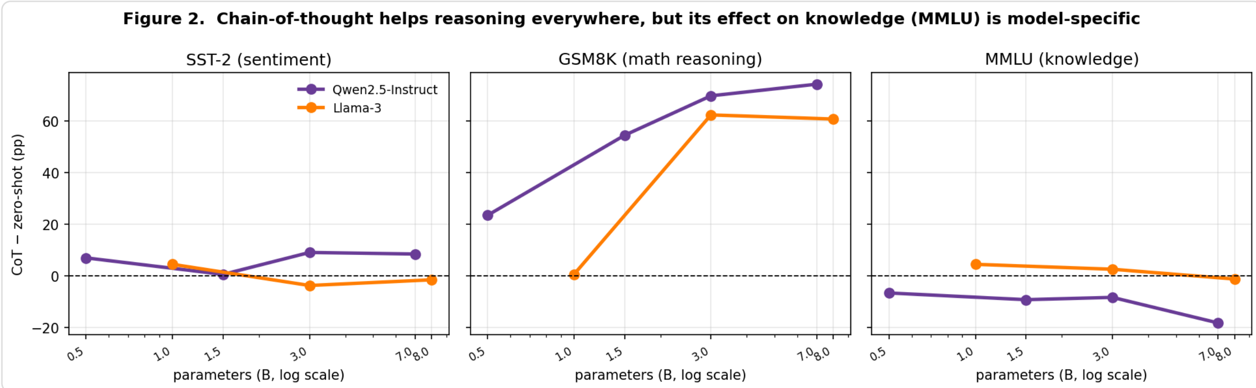

Figure 2. CoT minus zero-shot accuracy (pp) vs. model size. On GSM8K both families are large and positive. On MMLU the families diverge: Qwen (purple) drops increasingly below zero (CoT hurts, worst at 7B: −17pp), while Llama (orange) stays neutral-to-positive. On SST-2 both are small.

In Qwen2.5, the same CoT that dominates GSM8K *degrades* MMLU at every size — by 7.4pp (0.5B), 10.0pp (1.5B), 10.4pp (3B), and 17.0pp (7B) — with harm growing in scale. We interpret this as over-reasoning on multiple-choice items: free-form reasoning creates opportunities to argue away the correct option (cf. Tam et al., 2024).

This effect does not generalize across families. In Llama-3, CoT on MMLU is neutral-to-helpful (+6.0pp at 1B, +4.0pp at 3B, −2.0pp at 8B). We therefore report CoT's degradation of knowledge QA as a *model-family-dependent phenomenon, not a universal law*, while the broader claim — that prompting buys little on knowledge regardless of strategy (§4.2) — holds in both families. This divergence is exactly what a single-family study would have over-generalized.

Conversely, structured (JSON) output is *not* a universal penalty in either family — it is best-in-class on SST-2 (Qwen-7B: 95.3; Llama-8B: 91.7) and competitive on MMLU, and only clearly harmful on GSM8K, refining the "format tax" of Tam et al. (2024) to a reasoning-specific effect. We also observe that few-shot is markedly less reliable in Llama-3 than Qwen2.5 (e.g. GSM8K ≈1%, SST-2-1B 0.7%), consistent with strong family-specific sensitivity to exemplar formatting (Sclar et al., 2023; Lu et al., 2022).

4.4 The effects are not scoring artifacts

Continuous metric (GSM8K). Numeric-present tracks exact-match closely rather than diverging (Qwen-0.5B CoT 25.3 / 39.0; Qwen-7B CoT 88.0 / 91.0), and the median relative error of the committed answer is 0.000 for CoT at 1.5B and above. The low small-model scores are not an artifact of a discontinuous metric (Schaeffer et al., 2023).

Extraction audit (MMLU CoT). Re-extracting answers from full outputs, the extraction-failure rate is only 2–4%; scoring only cleanly-parsed answers, Qwen CoT still trails zero-shot by 6–16pp at every size. The CoT-hurts-knowledge effect in Qwen is real, not a parsing failure.

5. Discussion

The inversion is the point: prompting helps most exactly where one might expect it to help least. Reasoning tasks have latent capability that the right prompt (CoT) elicits — explaining both the large GSM8K exchange rate and CoT's scale-dependent gains. Knowledge tasks have no such latent slack: the fact is present or absent in the weights, so only parameters move the needle. A practical corollary: under a fixed budget, spend on prompting for reasoning workloads and on parameters for knowledge workloads — and be cautious applying CoT to multiple-choice knowledge QA, since whether it helps or hurts depends on the model family.

6. Limitations

Two seeds (42, 43) for both families; across seeds, condition accuracies differed by a mean of ~2.3pp (max 7.7pp), within the Wilson intervals, and every reported effect replicates. Greedy decoding only (no self-consistency or temperature sweep). Fixed hand-written exemplars; prompt-order and format sensitivity not ablated (Lu et al., 2022; Sclar et al., 2023), and Llama's weak few-shot results are likely an instance of this. 4-bit quantization is held constant but may interact with size; three benchmarks; English-only. GSM8K may probe brittle pattern-matching rather than reasoning (Mirzadeh et al., 2024), so the "reasoning" interpretation is a caveat. The Schaeffer (2023) concern is mitigated via the numeric-present metric but exact-match remains the headline; we report both.

7. Conclusion

Across a controlled two-family sweep, a good prompt is worth several model tiers on multi-step arithmetic but almost nothing on factual knowledge, and chain-of-thought — uniquely powerful on the former — can actively harm the latter, though whether it does so is model-specific. The lever you should pull depends on the task, and our exchange-rate framing makes that dependence quantitative.

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Appendix A. Prompt templates

Each prompt below is wrapped in the model's chat template (`apply_chat_template`, `add_generation_prompt=True`) before generation. `{...}` denotes the inserted example.

A.1 SST-2 (sentiment)

```
zero-shot:  Classify the sentiment of the following sentence as exactly
            'positive' or 'negative'.

Sentence: {sentence}
Sentiment:

few-shot:   Classify the sentiment as 'positive' or 'negative'.
            [3 exemplars: ("The film is a masterpiece of visual storytelling." -> posit
             ("I fell asleep twice. Complete waste of time." -> negative),
             ("A quietly moving portrait of grief and resilience." -> positive)]

Sentence: {sentence}
Sentiment:

CoT:        Identify the key sentiment signals in the sentence, reason through them,
            then classify as 'positive' or 'negative'. End your response with
            'Final answer: positive' or 'Final answer: negative'.
Sentence: {sentence}
Reasoning:

structured: Classify sentiment. Respond ONLY with valid JSON:
            {"sentiment": "positive"} or {"sentiment": "negative"}.
Sentence: {sentence}
JSON:
```

A.2 GSM8K (arithmetic)

```
zero-shot:  Solve the following math problem. Give only the final number as your
            answer.

Problem: {question}
Answer:

few-shot:   2 worked examples shown as Problem / Answer (final number only),
            then Problem: {question}
```

```
Answer:

CoT:        2 worked examples shown as Problem / Reasoning / Answer, then
            "Solve ... step by step. End with 'Answer: <number>'."

Problem: {question}
Reasoning:

structured: Solve this math problem. Respond ONLY with valid JSON:
            {"answer": "<number>"}.
Problem: {question}
JSON:
```

A.3 MMLU (knowledge, 4-way MC)

```
zero-shot:  Answer the following multiple choice question. Respond with only the
            letter A, B, C, or D.

Question: {q}
{A..D options}
Answer:

few-shot:   4 exemplars (answers spread across A/B/C/D to avoid position bias),
            then Question: {q}
{options}
Answer:

CoT:        "Think through each option carefully, then end your response with
            'Final answer: A' (or B, C, D)."
Question: {q}
{options}
Reasoning:

structured: Respond ONLY with valid JSON: {"answer": "A"} (or B, C, D).

Question: {q}
{options}
JSON:
```

Appendix B. Answer-extraction rules

Greedy decoding (temperature 0); answers parsed from raw model text as follows.

- **SST-2.** structured: parse JSON `sentiment` field. CoT: regex `final answer[:\s]+(positive|negative)`. Otherwise: scan the last 60 characters, then the whole output, for "positive"/"negative".

- **GSM8K.** structured: parse JSON `answer`. Otherwise: first match of `answer[:\s]+$?(number)`, then a trailing `= $number`, then the last number in the text (excluding bare years 2020-2026). Scored by numeric equality (tolerance 0.01).
- **MMLU.** structured: parse JSON `answer`. CoT: regex `final answer[:\s]+([A–D])`. Otherwise: a lone A-D letter on the last non-empty line, then the last A-D letter anywhere. Scored by exact letter match.

Extraction failures count as incorrect. An audit of MMLU-CoT on full (untruncated) outputs found a 2-4% extraction-failure rate, with the reported effects unchanged (§4.4).

Appendix C. Full per-condition results (95% Wilson CIs & McNemar tests)

Both ladders are pooled over seeds 42 and 43 (n = 600 per condition). McNemar tests are continuity-corrected and Holm-adjusted within each model×task, comparing each strategy to zero-shot on the same items.

C.1 Accuracy and 95% Wilson CI

Family	Model	Task	Strategy	n	Acc (%)	95% CI
Qwen2.5-Instruct	qwen-0.5b	sst2	zero-shot	600	60.2	[56.2, 64.0]
Qwen2.5-Instruct	qwen-0.5b	sst2	few-shot	600	85.5	[82.5, 88.1]
Qwen2.5-Instruct	qwen-0.5b	sst2	CoT	600	67.2	[63.3, 70.8]
Qwen2.5-Instruct	qwen-0.5b	sst2	structured	600	73.3	[69.7, 76.7]
Qwen2.5-Instruct	qwen-0.5b	gsm8k	zero-shot	600	3.2	[2.0, 4.9]
Qwen2.5-Instruct	qwen-0.5b	gsm8k	few-shot	600	3.8	[2.6, 5.7]
Qwen2.5-Instruct	qwen-0.5b	gsm8k	CoT	600	26.7	[23.3, 30.3]
Qwen2.5-Instruct	qwen-0.5b	gsm8k	structured	600	2.8	[1.8, 4.5]
Qwen2.5-Instruct	qwen-0.5b	mmlu	zero-shot	600	35.3	[31.6, 39.2]
Qwen2.5-Instruct	qwen-0.5b	mmlu	few-shot	600	27.5	[24.1, 31.2]
Qwen2.5-Instruct	qwen-0.5b	mmlu	CoT	600	28.7	[25.2, 32.4]
Qwen2.5-Instruct	qwen-0.5b	mmlu	structured	600	26.5	[23.1, 30.2]
Qwen2.5-Instruct	qwen-1.5b	sst2	zero-shot	600	84.2	[81.0, 86.9]
Qwen2.5-Instruct	qwen-1.5b	sst2	few-shot	600	91.0	[88.4, 93.0]
Qwen2.5-Instruct	qwen-1.5b	sst2	CoT	600	84.8	[81.7, 87.5]
Qwen2.5-Instruct	qwen-1.5b	sst2	structured	600	90.0	[87.3, 92.2]
Qwen2.5-Instruct	qwen-1.5b	gsm8k	zero-shot	600	3.2	[2.0, 4.9]
Qwen2.5-Instruct	qwen-1.5b	gsm8k	few-shot	600	7.5	[5.7, 9.9]
Qwen2.5-Instruct	qwen-1.5b	gsm8k	CoT	600	57.7	[53.7, 61.6]
Qwen2.5-Instruct	qwen-1.5b	gsm8k	structured	600	13.0	[10.5, 15.9]
Qwen2.5-Instruct	qwen-1.5b	mmlu	zero-shot	600	54.7	[50.7, 58.6]
Qwen2.5-Instruct	qwen-1.5b	mmlu	few-shot	600	54.8	[50.8, 58.8]
Qwen2.5-Instruct	qwen-1.5b	mmlu	CoT	600	45.5	[41.6, 49.5]
Qwen2.5-Instruct	qwen-1.5b	mmlu	structured	600	57.0	[53.0, 60.9]
Qwen2.5-Instruct	qwen-3b	sst2	zero-shot	600	79.7	[76.3, 82.7]
Qwen2.5-Instruct	qwen-3b	sst2	few-shot	600	81.0	[77.7, 83.9]
Qwen2.5-Instruct	qwen-3b	sst2	CoT	600	88.8	[86.1, 91.1]
Qwen2.5-Instruct	qwen-3b	sst2	structured	600	90.2	[87.5, 92.3]
Qwen2.5-Instruct	qwen-3b	gsm8k	zero-shot	600	2.5	[1.5, 4.1]
Qwen2.5-Instruct	qwen-3b	gsm8k	few-shot	600	2.0	[1.1, 3.5]
Qwen2.5-Instruct	qwen-3b	gsm8k	CoT	600	72.3	[68.6, 75.8]
Qwen2.5-Instruct	qwen-3b	gsm8k	structured	600	2.5	[1.5, 4.1]
Qwen2.5-Instruct	qwen-3b	mmlu	zero-shot	600	60.3	[56.4, 64.2]
Qwen2.5-Instruct	qwen-3b	mmlu	few-shot	600	59.2	[55.2, 63.0]
Qwen2.5-Instruct	qwen-3b	mmlu	CoT	600	52.0	[48.0, 56.0]
Qwen2.5-Instruct	qwen-3b	mmlu	structured	600	60.5	[56.5, 64.3]
Qwen2.5-Instruct	qwen-7b	sst2	zero-shot	600	84.5	[81.4, 87.2]
Qwen2.5-Instruct	qwen-7b	sst2	few-shot	600	90.3	[87.7, 92.4]

Qwen2.5-Instruct	qwen-7b	sst2	CoT	600	93.0	[90.7, 94.8]
Qwen2.5-Instruct	qwen-7b	sst2	structured	600	94.0	[91.8, 95.6]
Qwen2.5-Instruct	qwen-7b	gsm8k	zero-shot	600	14.7	[12.1, 17.7]
Qwen2.5-Instruct	qwen-7b	gsm8k	few-shot	600	2.3	[1.4, 3.9]
Qwen2.5-Instruct	qwen-7b	gsm8k	CoT	600	89.0	[86.2, 91.3]
Qwen2.5-Instruct	qwen-7b	gsm8k	structured	600	25.2	[21.9, 28.8]
Qwen2.5-Instruct	qwen-7b	mmlu	zero-shot	600	72.2	[68.4, 75.6]
Qwen2.5-Instruct	qwen-7b	mmlu	few-shot	600	72.8	[69.1, 76.2]
Qwen2.5-Instruct	qwen-7b	mmlu	CoT	600	54.0	[50.0, 57.9]
Qwen2.5-Instruct	qwen-7b	mmlu	structured	600	73.5	[69.8, 76.9]
Llama-3	llama-1b	sst2	zero-shot	600	64.8	[60.9, 68.5]
Llama-3	llama-1b	sst2	few-shot	600	0.3	[0.1, 1.2]
Llama-3	llama-1b	sst2	CoT	600	69.3	[65.5, 72.9]
Llama-3	llama-1b	sst2	structured	600	73.2	[69.5, 76.6]
Llama-3	llama-1b	gsm8k	zero-shot	600	5.7	[4.1, 7.8]
Llama-3	llama-1b	gsm8k	few-shot	600	2.0	[1.1, 3.5]
Llama-3	llama-1b	gsm8k	CoT	600	6.3	[4.6, 8.6]
Llama-3	llama-1b	gsm8k	structured	600	5.3	[3.8, 7.4]
Llama-3	llama-1b	mmlu	zero-shot	600	31.2	[27.6, 35.0]
Llama-3	llama-1b	mmlu	few-shot	600	3.5	[2.3, 5.3]
Llama-3	llama-1b	mmlu	CoT	600	35.7	[31.9, 39.6]
Llama-3	llama-1b	mmlu	structured	600	42.2	[38.3, 46.2]
Llama-3	llama-3b	sst2	zero-shot	600	88.7	[85.9, 91.0]
Llama-3	llama-3b	sst2	few-shot	600	57.5	[53.5, 61.4]
Llama-3	llama-3b	sst2	CoT	600	85.0	[81.9, 87.6]
Llama-3	llama-3b	sst2	structured	600	89.7	[87.0, 91.9]
Llama-3	llama-3b	gsm8k	zero-shot	600	10.8	[8.6, 13.6]
Llama-3	llama-3b	gsm8k	few-shot	600	1.3	[0.7, 2.6]
Llama-3	llama-3b	gsm8k	CoT	600	73.2	[69.5, 76.6]
Llama-3	llama-3b	gsm8k	structured	600	9.8	[7.7, 12.5]
Llama-3	llama-3b	mmlu	zero-shot	600	52.7	[48.7, 56.6]
Llama-3	llama-3b	mmlu	few-shot	600	34.0	[30.3, 37.9]
Llama-3	llama-3b	mmlu	CoT	600	55.3	[51.3, 59.3]
Llama-3	llama-3b	mmlu	structured	600	56.7	[52.7, 60.6]
Llama-3	llama-8b	sst2	zero-shot	600	92.0	[89.6, 93.9]
Llama-3	llama-8b	sst2	few-shot	600	85.0	[81.9, 87.6]
Llama-3	llama-8b	sst2	CoT	600	90.5	[87.9, 92.6]
Llama-3	llama-8b	sst2	structured	600	92.2	[89.7, 94.1]
Llama-3	llama-8b	gsm8k	zero-shot	600	20.7	[17.6, 24.1]
Llama-3	llama-8b	gsm8k	few-shot	600	1.3	[0.7, 2.6]

Llama-3	llama-8b	gsm8k	CoT	600	81.5	[78.2, 84.4]
Llama-3	llama-8b	gsm8k	structured	600	9.7	[7.6, 12.3]
Llama-3	llama-8b	mmlu	zero-shot	600	62.5	[58.6, 66.3]
Llama-3	llama-8b	mmlu	few-shot	600	56.8	[52.8, 60.7]
Llama-3	llama-8b	mmlu	CoT	600	61.3	[57.4, 65.1]
Llama-3	llama-8b	mmlu	structured	600	64.0	[60.1, 67.7]

C.2 McNemar tests vs. zero-shot (Holm-adjusted)

Family	Model	Task	Comparison	p (adj)	
Qwen2.5-Instruct	qwen-0.5b	sst2	zero-shot vs few-shot	0.0000	***
Qwen2.5-Instruct	qwen-0.5b	sst2	zero-shot vs CoT	0.0006	***
Qwen2.5-Instruct	qwen-0.5b	sst2	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-0.5b	gsm8k	zero-shot vs few-shot	1.0000	ns
Qwen2.5-Instruct	qwen-0.5b	gsm8k	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-0.5b	gsm8k	zero-shot vs structured	1.0000	ns
Qwen2.5-Instruct	qwen-0.5b	mmlu	zero-shot vs few-shot	0.0002	***
Qwen2.5-Instruct	qwen-0.5b	mmlu	zero-shot vs CoT	0.0015	**
Qwen2.5-Instruct	qwen-0.5b	mmlu	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	sst2	zero-shot vs few-shot	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	sst2	zero-shot vs CoT	0.7373	ns
Qwen2.5-Instruct	qwen-1.5b	sst2	zero-shot vs structured	0.0002	***
Qwen2.5-Instruct	qwen-1.5b	gsm8k	zero-shot vs few-shot	0.0010	**
Qwen2.5-Instruct	qwen-1.5b	gsm8k	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	gsm8k	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	mmlu	zero-shot vs few-shot	1.0000	ns
Qwen2.5-Instruct	qwen-1.5b	mmlu	zero-shot vs CoT	0.0004	***
Qwen2.5-Instruct	qwen-1.5b	mmlu	zero-shot vs structured	0.2922	ns
Qwen2.5-Instruct	qwen-3b	sst2	zero-shot vs few-shot	0.5045	ns
Qwen2.5-Instruct	qwen-3b	sst2	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-3b	sst2	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-3b	gsm8k	zero-shot vs few-shot	1.0000	ns
Qwen2.5-Instruct	qwen-3b	gsm8k	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-3b	gsm8k	zero-shot vs structured	1.0000	ns
Qwen2.5-Instruct	qwen-3b	mmlu	zero-shot vs few-shot	1.0000	ns
Qwen2.5-Instruct	qwen-3b	mmlu	zero-shot vs CoT	0.0035	**
Qwen2.5-Instruct	qwen-3b	mmlu	zero-shot vs structured	1.0000	ns
Qwen2.5-Instruct	qwen-7b	sst2	zero-shot vs few-shot	0.0002	***
Qwen2.5-Instruct	qwen-7b	sst2	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-7b	sst2	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-7b	gsm8k	zero-shot vs few-shot	0.0000	***
Qwen2.5-Instruct	qwen-7b	gsm8k	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-7b	gsm8k	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-7b	mmlu	zero-shot vs few-shot	0.6774	ns
Qwen2.5-Instruct	qwen-7b	mmlu	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-7b	mmlu	zero-shot vs structured	0.5123	ns
Llama-3	llama-1b	sst2	zero-shot vs few-shot	0.0000	***
Llama-3	llama-1b	sst2	zero-shot vs CoT	0.0653	ns

Llama-3	llama-1b	sst2	zero-shot vs structured	0.0002	***
Llama-3	llama-1b	gsm8k	zero-shot vs few-shot	0.0059	**
Llama-3	llama-1b	gsm8k	zero-shot vs CoT	1.0000	ns
Llama-3	llama-1b	gsm8k	zero-shot vs structured	1.0000	ns
Llama-3	llama-1b	mmlu	zero-shot vs few-shot	0.0000	***
Llama-3	llama-1b	mmlu	zero-shot vs CoT	0.0844	ns
Llama-3	llama-1b	mmlu	zero-shot vs structured	0.0001	***
Llama-3	llama-3b	sst2	zero-shot vs few-shot	0.0000	***
Llama-3	llama-3b	sst2	zero-shot vs CoT	0.0293	*
Llama-3	llama-3b	sst2	zero-shot vs structured	0.3074	ns
Llama-3	llama-3b	gsm8k	zero-shot vs few-shot	0.0000	***
Llama-3	llama-3b	gsm8k	zero-shot vs CoT	0.0000	***
Llama-3	llama-3b	gsm8k	zero-shot vs structured	0.5982	ns
Llama-3	llama-3b	mmlu	zero-shot vs few-shot	0.0000	***
Llama-3	llama-3b	mmlu	zero-shot vs CoT	0.2415	ns
Llama-3	llama-3b	mmlu	zero-shot vs structured	0.0482	*
Llama-3	llama-8b	sst2	zero-shot vs few-shot	0.0000	***
Llama-3	llama-8b	sst2	zero-shot vs CoT	0.4230	ns
Llama-3	llama-8b	sst2	zero-shot vs structured	1.0000	ns
Llama-3	llama-8b	gsm8k	zero-shot vs few-shot	0.0000	***
Llama-3	llama-8b	gsm8k	zero-shot vs CoT	0.0000	***
Llama-3	llama-8b	gsm8k	zero-shot vs structured	0.0000	***
Llama-3	llama-8b	mmlu	zero-shot vs few-shot	0.0023	**
Llama-3	llama-8b	mmlu	zero-shot vs CoT	0.7598	ns
Llama-3	llama-8b	mmlu	zero-shot vs structured	0.7598	ns

Significance: *** p<.001, ** p<.01, * p<.05, ns = not significant.

Appendix D. Compute and reproducibility

All experiments ran on a single Apple M4 laptop (16 GB unified memory) via MLX, one model resident at a time. Models are the 4-bit mlx-community quantizations of Qwen2.5-Instruct (0.5/1.5/3/7B) and Llama-3 (Llama-3.2-1B/3B, Meta-Llama-3.1-8B). Decoding is greedy (temperature 0). Max new tokens per condition: 8 (MMLU), 16 (SST-2), 48 (GSM8K non-CoT), 256 (SST-2/MMLU CoT), 512 (GSM8K CoT). n = 300 examples per condition; MMLU is stratified across all 57 subjects. Primary results use seed 42; a second seed (43) is reported for variance where available. The full harness (`run_experiments_local.py`), prompt builder, extraction code, prompt templates, and per-example results are released at github.com/ssamalsamir/prompting-vs-model-scaling — `--aggregate` reproduces all tables from the saved per-example CSVs.

> **Status.** Author/school filled in; all 13 citations verified against arXiv (one correction applied — arXiv:2212.01907 is by San Joaquin & Haroen, not Shaikh); Appendices A–D built; figures generated from the data. Both families were run at two seeds (42, 43) — cross-seed accuracy differed by ~2.3pp on average (7.7pp max), within the Wilson intervals, and Appendix C pools both seeds (n = 600). Per the author's choice, no external mentor review was sought; one is still recommended before submitting to JEI / arXiv.